Kothagudem Coalfields

🏭 Introduction

The **Kothagudem Coalfields** are an important coal mining region in southern India, located in Telangana. These coalfields are operated by Singareni Collieries Company Limited, a government-owned enterprise jointly controlled by the Government of Telangana and the Government of India.

Kothagudem coalfields are known for their **non-coking (thermal) coal**, which is mainly used for power generation and industrial purposes. Mining in this region includes both **underground and opencast methods**, making it significant from a mining engineering perspective. The coalfields play a vital role in supplying coal to thermal power plants in southern India.

🌍 Geographical Overview – Kothagudem Coalfields

The Kothagudem coalfields are located in the **Bhadradri Kothagudem district** of Telangana and form part of the **Godavari Valley coalfields**.

📍 Location & Setting

- Situated in eastern Telangana near the Andhra Pradesh border
- Lies within the **Godavari Valley coal basin**
- Well connected by rail and road networks for coal transportation

🌄 Physical & Geological Features

- Part of the **Gondwana coalfields**, containing significant coal deposits
- Features **moderately thick coal seams**, suitable for both underground and opencast mining
- Coal type: **non-coking thermal coal**

🌊 Nearby Natural Features

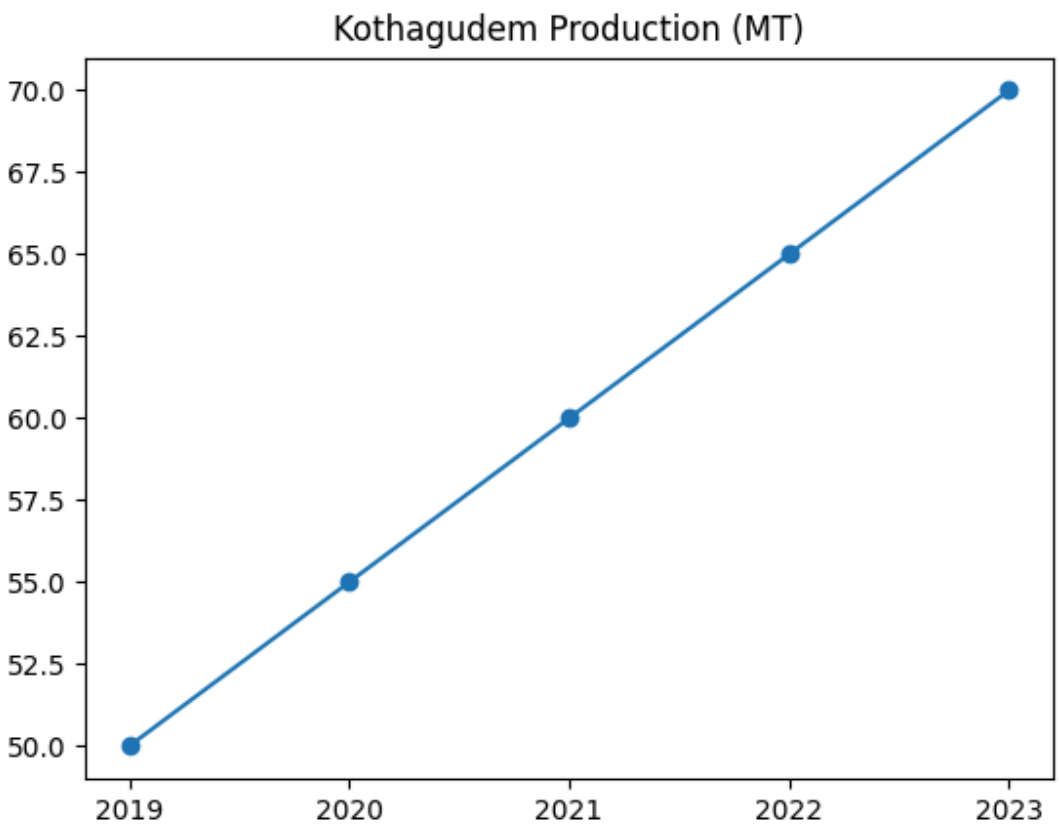
- Located near the Godavari River basin
- Surrounded by forested and hilly terrain

🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)

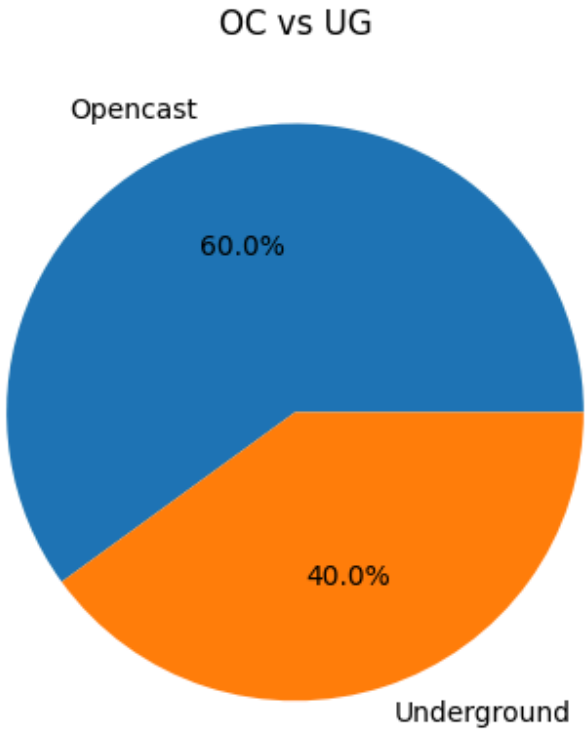
- Moderate monsoon rainfall
- Mild winters

📊 1. Production Trend (MT)



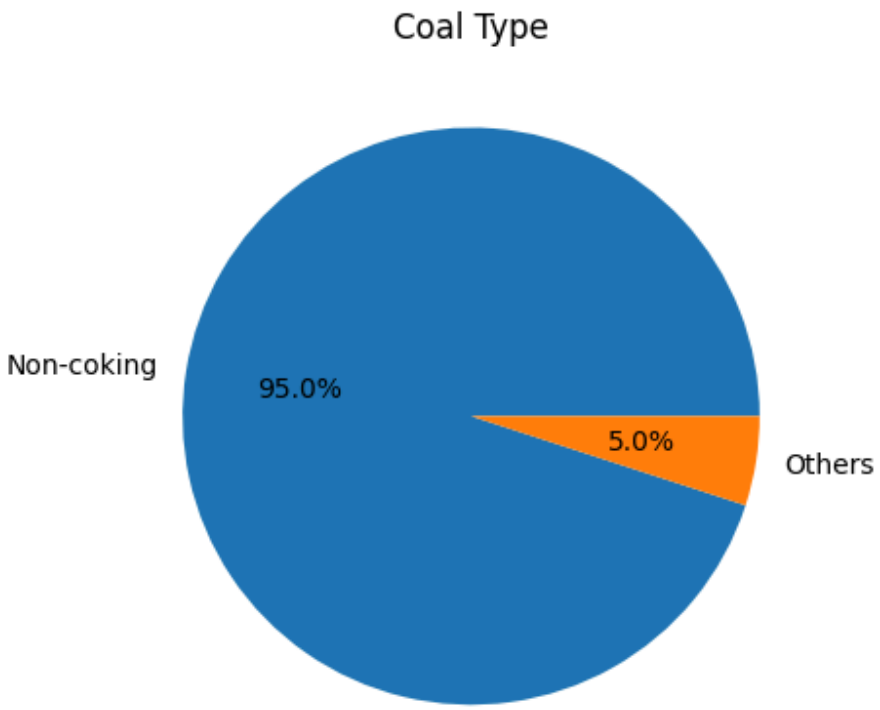
This chart shows the steady increase in coal production in Kothagudem under Singareni Collieries Company Limited.
The growth reflects expansion of both opencast and underground mines.
Significance: Useful for production forecasting and evaluating regional coal output.

📊 2. OC vs UG Mining Share



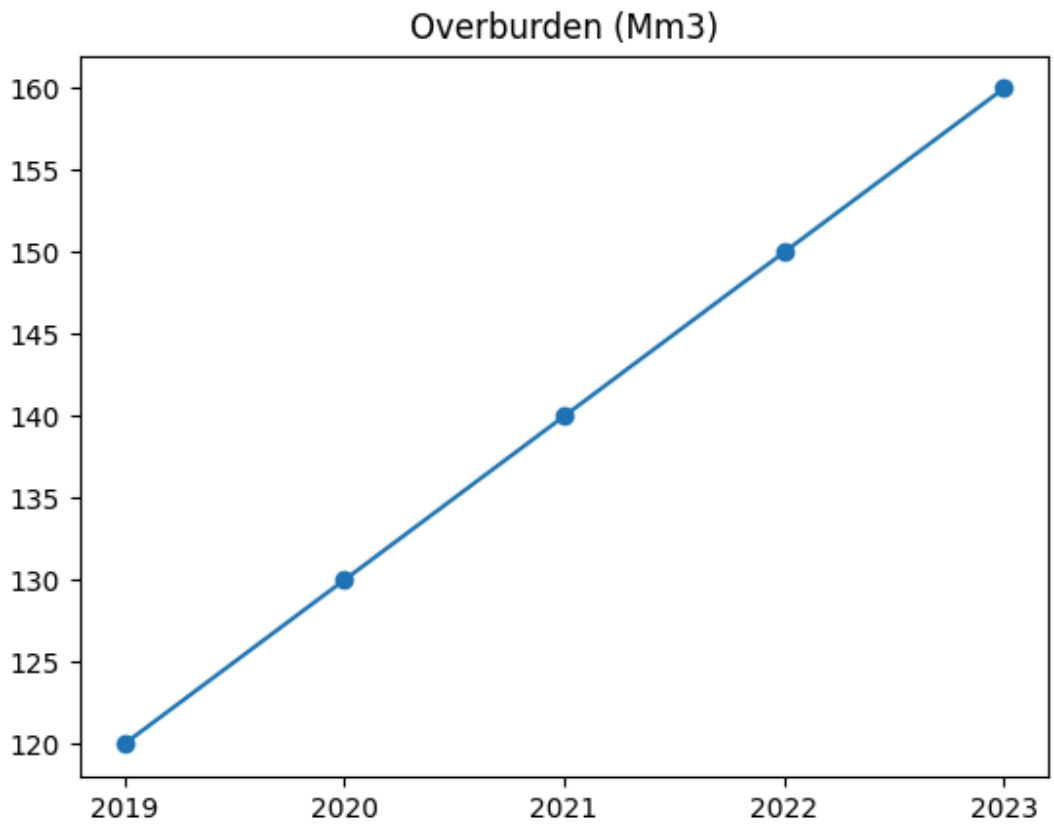
Represents the contribution of opencast and underground mining.
Both methods are significant, with opencast slightly dominating.
Significance: Helps compare productivity, cost, and efficiency of mining methods.

📊 3. Coal Type Distribution



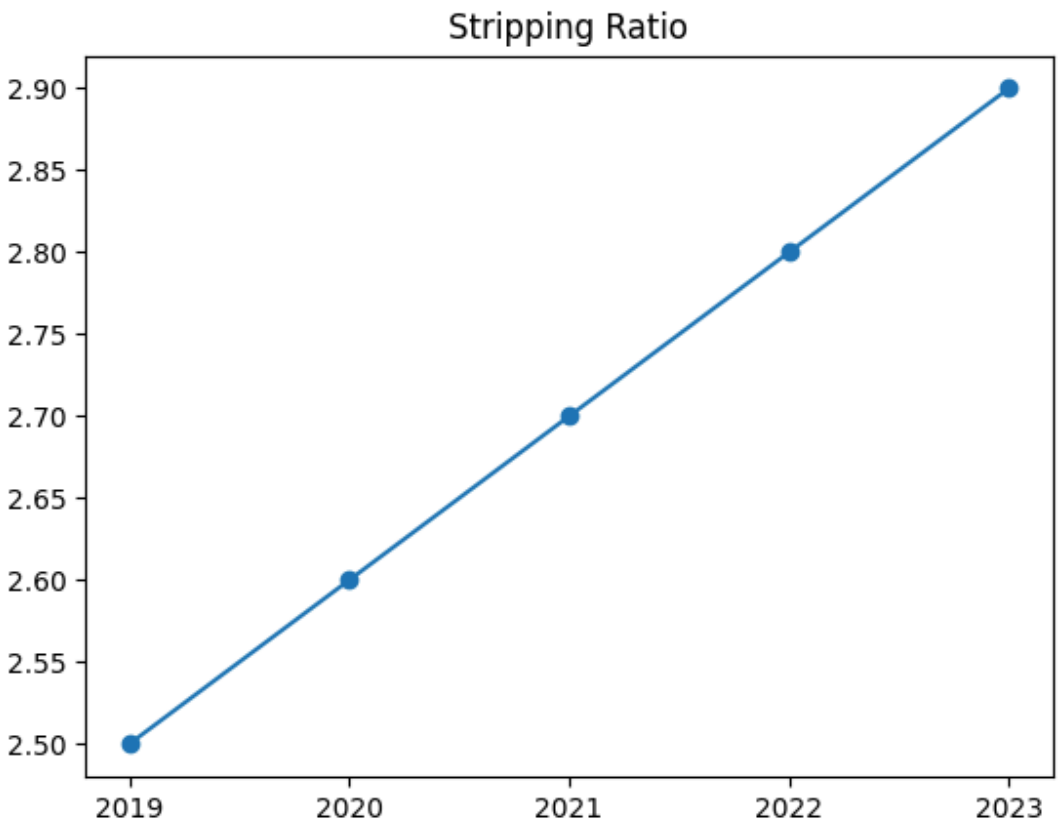
Shows dominance of non-coking (thermal) coal.
This coal is mainly used in power generation.
Significance: Determines industrial usage and highlights role in energy production.

📊 4. Overburden Removal (Mm³)



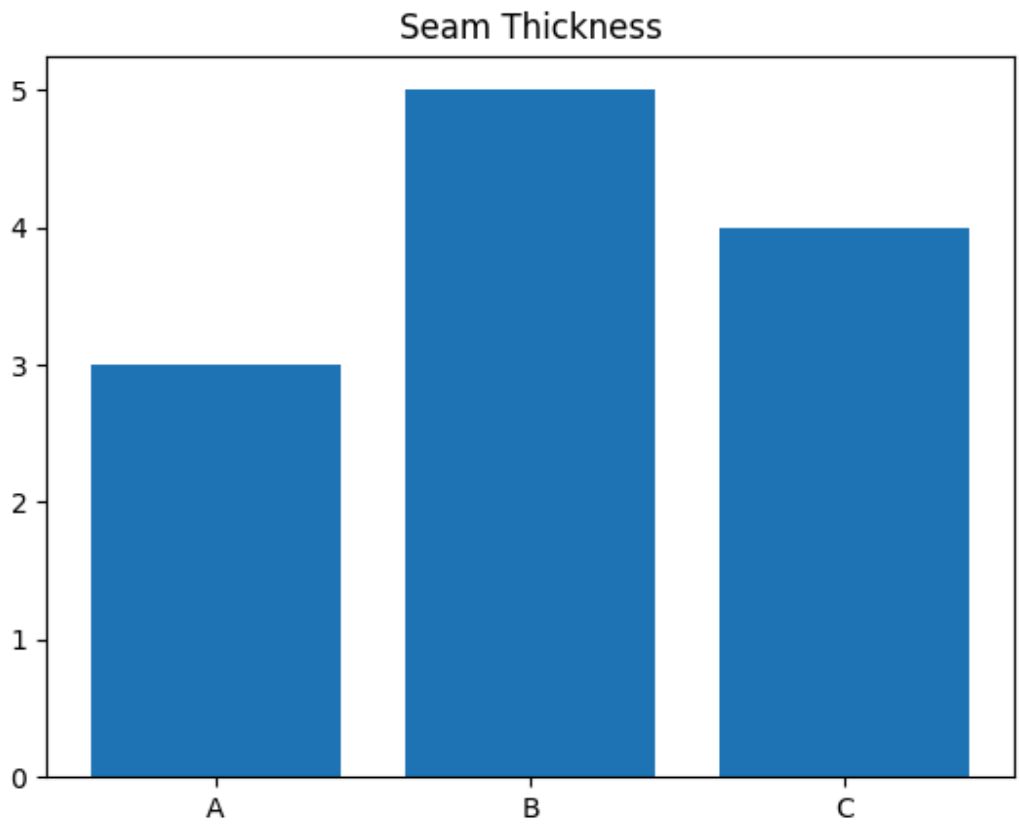
Displays the volume of overburden removed over time.
Increasing values indicate expansion of opencast mining.
Significance: Important for dump planning, equipment requirement, and cost estimation.

📊 5. Stripping Ratio Trend



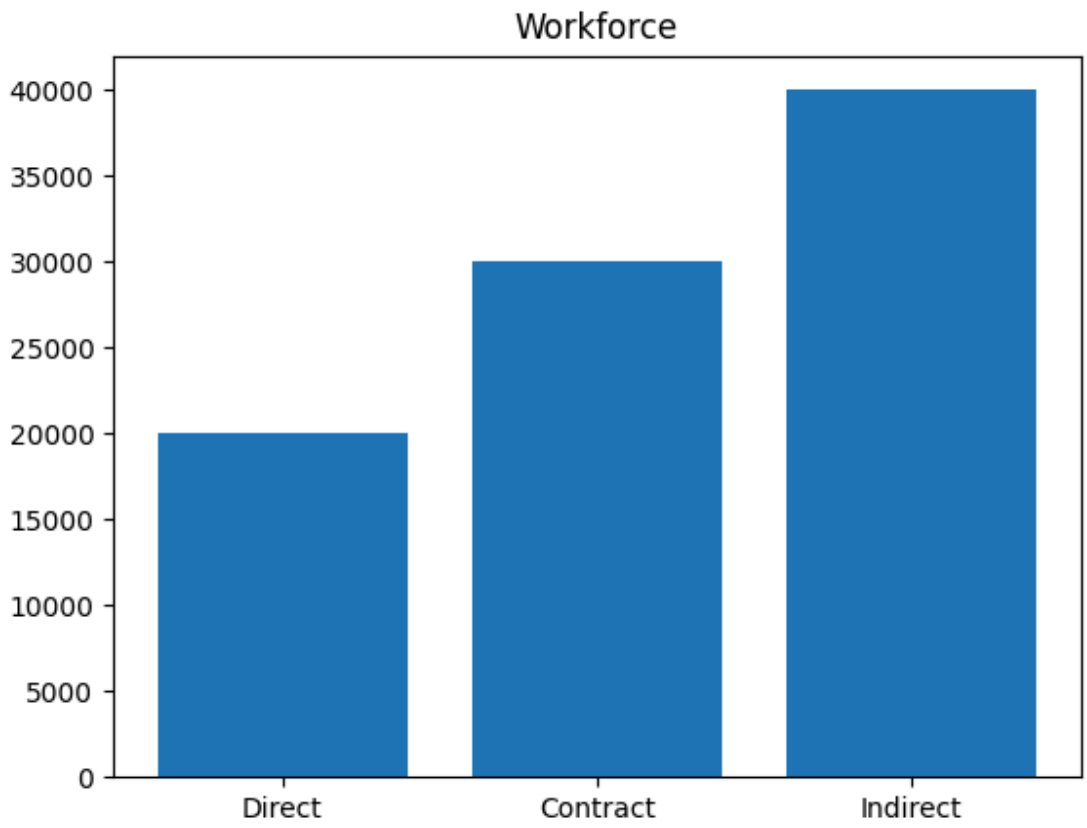
Shows gradual increase in stripping ratio.
Indicates deeper mining and rising operational cost.
Significance: Key parameter for economic feasibility and mine planning.

📊 6. Seam Thickness



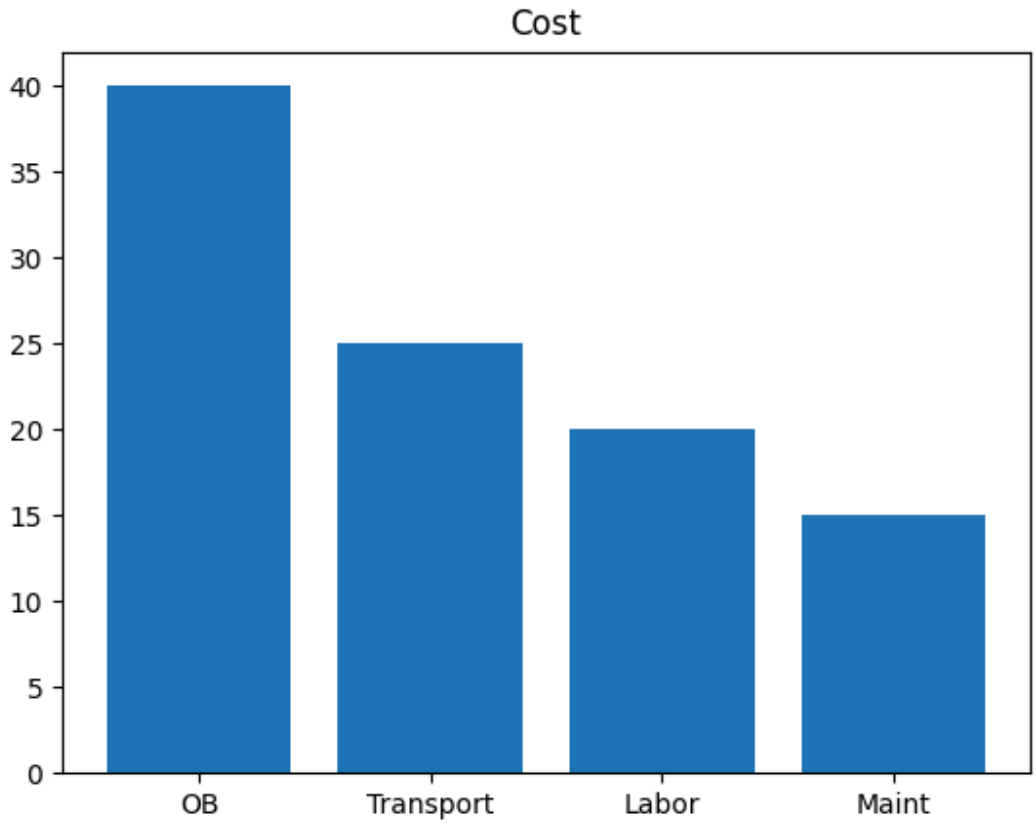
Represents thickness of coal seams in the coalfield.
Moderate seam thickness supports both UG and OC mining.
Significance: Helps in mine design, method selection, and equipment planning.

📊 7. Workforce Distribution



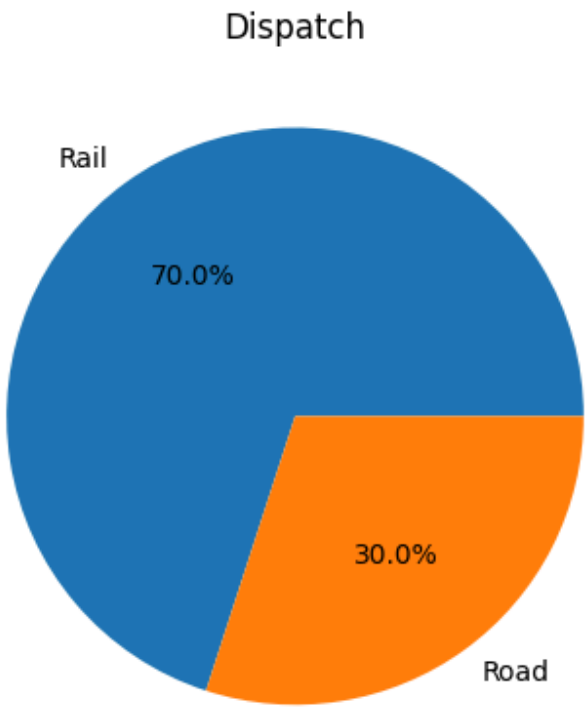
Displays manpower distribution across different employment categories.
Mining generates significant employment in the region.
Significance: Useful for manpower planning and socio-economic impact analysis.

📊 8. Cost Distribution



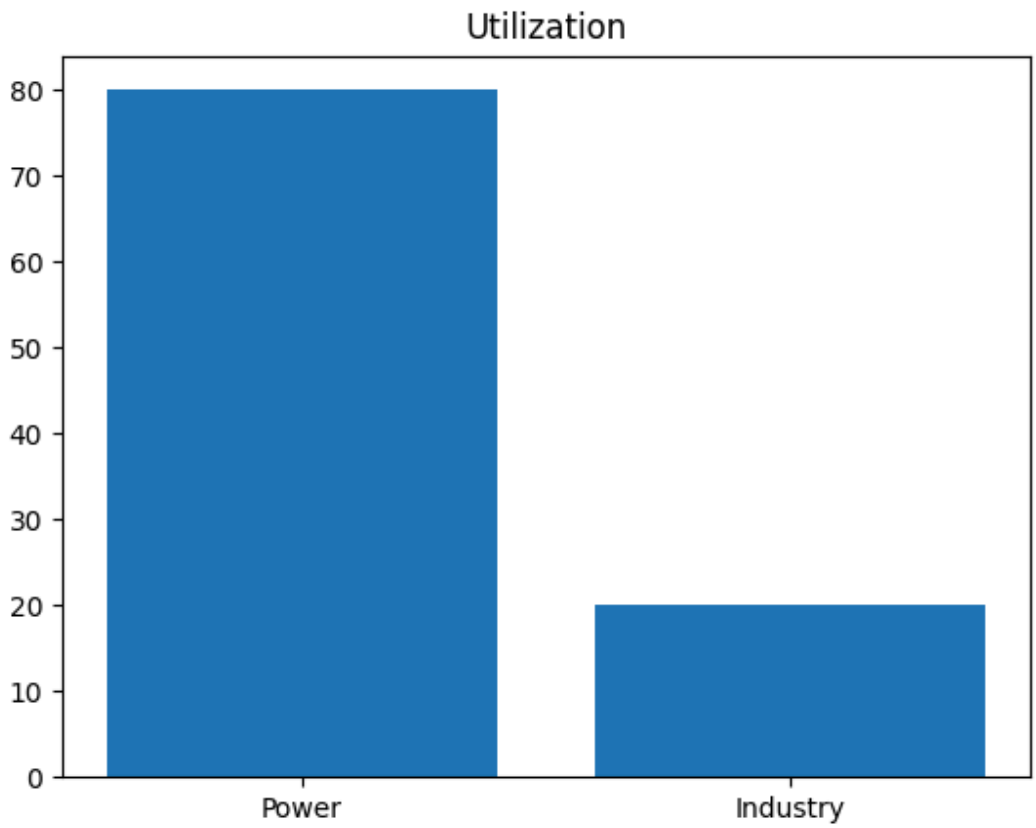
Shows breakdown of mining costs into major components.
Overburden removal forms the largest share.
Significance: Helps in cost optimization and improving operational efficiency.

📊 9. Coal Dispatch Mode



Represents transportation of coal via rail and road.
Rail transport dominates due to bulk handling efficiency.
Significance: Important for logistics planning and reducing transportation cost.

📊 10. Coal Utilization



Shows coal usage mainly in power generation.
Power sector consumes the majority of output.
Significance: Highlights the importance of Kothagudem coalfields in regional energy supply.